

CST8508 – Machine Vision – Assignment 2

Introduction

This is the second assignment that is aimed at giving you hands on exposure to training deep-learning models for computer vision tasks. We will be utilizing the [openmmlab](#) project for assignments.

Learning outcomes

For this assignment, you will be training an object detection model using [mmdetection](#). The main

learning outcomes from this assignment are:

- Now that you have understood that config file structure of openmmlab from assignment1, it should be easier for you to setup and train an object detection model using mmdetection.
- Converting dataset annotations between different formats
- Empirical evaluation of performance of different object detection models by analyzing their ROC curve and area under curve metrics.

Instructions

- You must first download the dataset. We will be using the [Oxford-IIIT Pet Dataset](#) for this assignment.
- You will then prepare the dataset such that it is compatible with mmdetection structure for training object detection models. For this dataset, I will ask you to prepare it in the coco format so that you do not have to create any annotation files. You can find more information about the coco format [here](#). Successfully preparing the dataset in this format is worth **40%** of the assignment grade. Another useful link is [here](#).
- Setup mmdetection in your environment. You can refer to the documentation [here](#). Make sure to run the “Verify the installation section” in the documentation to ensure you have a working environment before you move on.
- Now that you have a good understanding of the config files, I want you to train TOOD and VFNET on the dataset. The successful training of two models is worth **20%** of the assignment grade. Do not train for more than 20 epochs.
- Evaluate your models by plotting the ROC curve and calculating the area under the curve of each model. This evaluation is worth **30%** of the assignment grade.
- Lessons learned – write a summary of the challenges you faced with this assignment, how you solved them, lessons learned, and any other thoughts you have. This is worth **10%** of your assignment grade. You should also elaborate on how you decided on which checkpoint to select for comparing each model, and your final interpretation over which model performs better than the other.

What to submit on Brightspace

- Generated training log files
- All python code written
- Config files used
- Report showing your evaluation analysis and lessons learned.

OR

If you encounter difficulties with OpenMMLab and MMDetection for this assignment, you can select any framework and choose two models of your preference to train and compare.